Sustainable Forest Management

Summary

Most FGD respondents indicated that deforestation, charcoal production and fire frequency were decreasing and wildlife populations were increasing with their CFAs, and that natural resource management was improving within their CFAs. In addition, most FGD groups indicated that both deforestation and charcoal production were decreasing outside their CFA, suggesting there is no leakage or potentially a positive leakage effect. Community female FGDs tended to have a more positive view of the impact of CFM on SFM compared to the other groups, which is noteworthy given it is primarily the women who harvest forest products, but who often do not have voice in traditional natural resource management. However, it is also important to note that for a minority of CFAs both FGD groups and individual interviewees suggested the situation was worsening. Moreover, the governance index was a significant predictor of our SFM performance index. Very few CFMGs were practising restoration or establishing plantations. Existing CFAs should be expanded and new ones designed so that they incorporate degraded areas and contribute to national restoration and plantation targets. While most community groups felt SFM indicators were improving under CFM, it is important to identify those CFAs that are failing and address the causes, which our results suggest are likely to be linked to poor governance.

Key conclusions

According to the perceptions expressed by the FGDs, CFM in Zambia is improving SFM across a substantial majority of surveyed communities, with positive spillover effects in the surrounding landscape and encouraging indications of women's engagement in forest governance. However, the survey identified a minority of CFAs in which the situation was worsening and the governance index was a significant predictor of SFM performance. The identified link between governance quality and SFM performance is critical and underscores the need for improving governance across the CFM portfolio. Zambia could also benefit from a systematic approach to restoration and plantation through CFM.

Introduction

In this chapter, we consider the effect of community forestry on indicators of SFM. The information analysed is based entirely only on the responses of interviewees to our survey tool. In the next chapter, we will consider the effects of CFA on SMF as measured through remote sensing derived metrics.

We hypothesised that, as a result of CFM, communities would report significant reductions in deforestation, fire frequency, and charcoal production, increases in wildlife populations, and forest recovery. We also tested the hypothesis from the governance chapter that CFMGs with better governance would report stronger improvements in forest conditions than those with weaker governance.

Deforestation

We asked interviewees whether CFM has reduced deforestation inside their CFA and whether or not deforestation has increased outside their CFA.

Deforestation inside the community forest

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A majority of groups reported either a moderate or significant reduction in deforestation inside the CFA (Figure 1). Moreover, these answers were consistent across the FGD groups. There were no significant differences in response among FGD types (Table 1). The quality of governance, as measured by the governance index we generated in the previous chapter, was a highly significant predictor of the perceived decrease in deforestation within the CFA (Table 1).

Among the individual interviewees (Table 2), a majority of interviewees also indicated that CFM had reduced deforestation inside the CFA. The differences among the interviewee roles were not significant (Table 3). However, there was a significant effect of gender. Male interviewees were significantly more likely to state that CFM had reduced deforestation (Table 3).

Deforestation outside the community forest

Sometimes when activities are proscribed in one area they may be displaced other areas; an effect known as leakage. To investigate possible leakage effects, we also asked interviewees whether CFM had led to an increase in deforestation outside their community forest (Figure 2).

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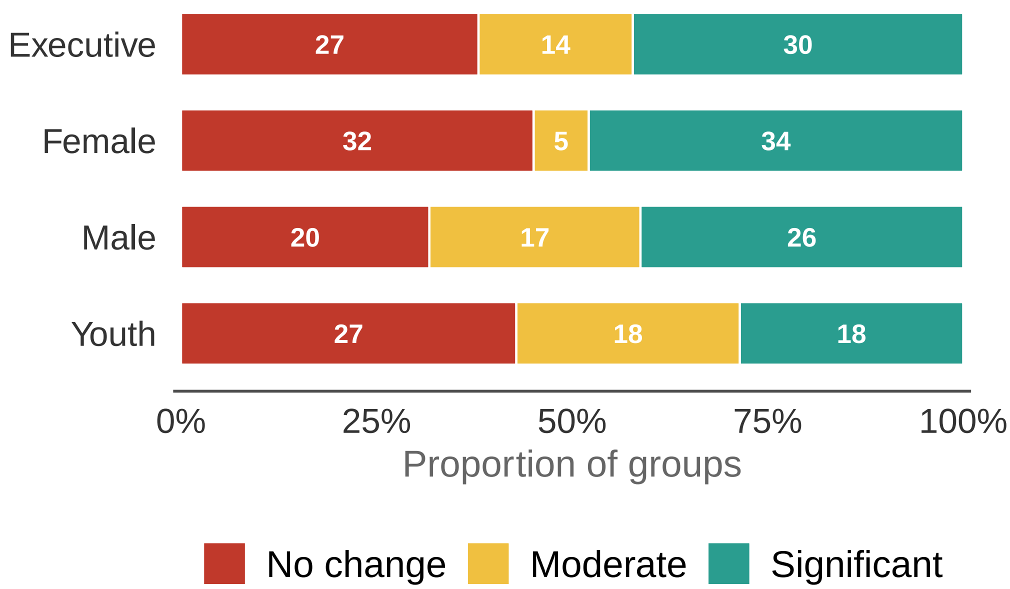

Figure 1: 'Has CFM reduced deforestation inside the community forest?' The length of each bar segment represents the proportion of FGD groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 1: Results of the model examining the effect of FGD type and governance index on perceived reduction in deforestation inside the CFA.

Model results for perceived reduction in deforestation inside the CFA — FGD type and governance index

We used a CLMM model with random intercepts for province and CFA within province. Reference = Executive. n = 254.

Factor	Estimate	SE	*P* value
Female	-0.133	0.354	0.7081
Male	0.137	0.351	0.6971
Youth	-0.412	0.347	0.2355
Governance index	0.291	0.071	0.0000

Table 2: Perceived reduction in deforestation inside the community forest by individual interviewee role.

Perceived reduction in deforestation inside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	No change	Moderate	Significant
CFMG executive member	60	17%	23%	60%
Community member	29	28%	24%	48%
Traditional leader	70	50%	17%	33%
District Forestry Officer	51	22%	51%	27%
Private enterprise rep	30	20%	40%	40%

Table 3: Results of the model examining the effect of interviewee role, gender and age on perceived reduction in deforestation inside the CFA.

Model results for perceived reduction in deforestation inside the CFA — interviewee role, gender and age

We used a CLMM model with random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 240.

Factor	Estimate	SE	P value
CFMG executive member	0.075	0.512	0.8841
Traditional leader	-0.943	0.490	0.0542
DFO	-0.701	0.493	0.1552
Private enterprise rep	-0.225	0.545	0.6803
Gender: male	0.776	0.362	0.0320
Age	-0.016	0.012	0.1878

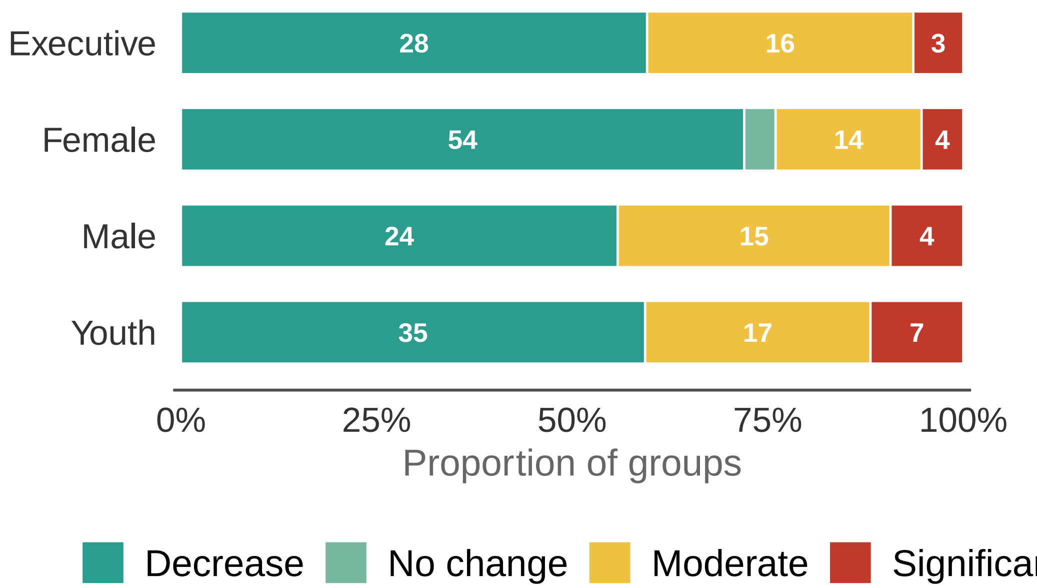

Figure 2: 'Has CFM increased deforestation outside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 4: Results of the model examining the effect of FGD type and governance index on perceived increase in deforestation outside the CFA.

Model results for perceived increase in deforestation outside the CFA — FGD type and governance index

We used a CLMM model with random intercepts for province and CFA within province. Reference = Executive. n = 209.

Factor	Estimate	SE	*P* value
Female	0.975	0.441	0.0270
Male	-0.083	0.461	0.8577
Youth	0.052	0.426	0.9022
Governance index	0.073	0.088	0.4095

Contrary to the expectations of leakage, a majority of the FGD groups responded that CFM had actually reduced deforestation outside their CFA. This result was consistent across FGD groups and the effect of governance was not significant (Table 4; (Table 5).

Among the individual interviewees (Table 6), the individuals from the communities agreed that deforestation outside their CFA had decreased.

The effects of the gender and age of the interviewee were not a significant determinant of their response. However, the differences in opinion between the CFMG executive members and DFOs, and between traditional leaders and DFOs, were significant (Table 7, Table 8).

Charcoal production

We asked interviewees whether CFM had reduced charcoal production inside their CFA and whether it had increased immediately outside their CFA.

Charcoal production inside the community forest

Almost all the FGD groups agreed that charcoal making inside their CFA had decreased, with most indicating that charcoal making had decreased significantly (Figure 3). The small differences in responses among FGD groups were not significant (Table 9). The effect of governance index was also not significant, likely reflecting the lack of variance in responses.

Likewise, among the individual interviewees (Table 10), almost all interviewees agreed that there had been a substantial decrease in charcoal making inside their CFA. Again, this result was consistent across interviewee roles (there were no significant differences among roles) (Table 11).

Table 5: Pairwise comparisons for perceived increase in deforestation outside the CFA — FGD type.

Pairwise comparisons for perceived increase in deforestation outside the CFA — FGD type

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	-0.975	0.441	-2.211	0.1202
Executive - Male	0.083	0.461	0.179	0.9980
Executive - Youth	-0.052	0.426	-0.123	0.9993
Female - Male	1.058	0.464	2.281	0.1023
Female - Youth	0.923	0.419	2.202	0.1226
Male - Youth	-0.135	0.447	-0.302	0.9904

Table 6: Perceived increase in deforestation outside the community forest by individual interviewee role.

Perceived increase in deforestation outside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	Moderate	Significant	No change
CFMG executive member	56	75%	16%	9%	0%
Community member	28	50%	39%	11%	0%
Traditional leader	68	72%	21%	6%	1%
District Forestry Officer	40	25%	52%	20%	2%
Private enterprise rep	23	39%	48%	13%	0%

Table 7: Results of the model examining the effect of interviewee role, gender and age on perceived increase in deforestation outside the CFA.

Model results for perceived increase in deforestation outside the CFA — interviewee role, gender and age

We used a CLMM model with random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 215.

Factor	Estimate	SE	P value
CFMG executive member	1.196	0.518	0.0211
Traditional leader	1.132	0.502	0.0241
DFO	-0.833	0.508	0.1010
Private enterprise rep	-0.396	0.582	0.4968
Gender: male	-0.271	0.398	0.4954
Age	-0.014	0.013	0.2738

Table 8: Pairwise comparisons for perceived increase in deforestation outside the CFA — interviewee role.

Pairwise comparisons for perceived increase in deforestation outside the CFA — interviewee role

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	P value
Community member - CFMG exec member	-1.196	0.518	-2.306	0.1427
Community member - Trad leader	-1.132	0.502	-2.255	0.1598
Community member - dfo	0.833	0.508	1.640	0.4718
Community member - NGO rep	0.396	0.582	0.680	0.9609
CFMG exec member - Trad leader	0.064	0.476	0.134	0.9999
CFMG exec member - dfo	2.028	0.467	4.347	0.0001
CFMG exec member - NGO rep	1.591	0.572	2.783	0.0429
Trad leader - dfo	1.964	0.482	4.077	0.0004
Trad leader - NGO rep	1.527	0.569	2.685	0.0562
dfo - NGO rep	-0.437	0.523	-0.836	0.9194

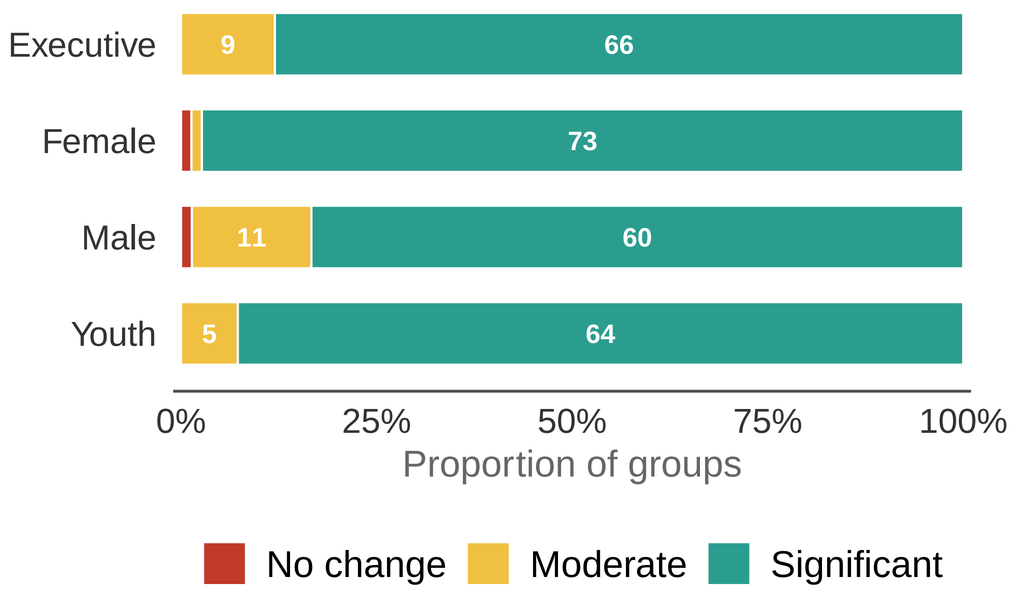

Figure 3: 'Has CFM reduced charcoal production inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 9: Results of the model examining the effect of FGD type and governance index on perceived reduction in charcoal production inside the CFA.

Model results for perceived reduction in charcoal production inside the CFA — FGD type and governance index

We used a CLMM model with random intercepts for province and CFA within province. Reference = Executive. n = 272.

Factor	Estimate	SE	P value
Female	1.525	0.826	0.0647
Male	0.100	0.569	0.8598
Youth	0.478	0.615	0.4366
Governance index	-0.056	0.135	0.6811

Table 10: Perceived reduction in charcoal production inside the community forest by individual interviewee role.

Perceived reduction in charcoal production inside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Moderate	Significant	No change
CFMG executive member	63	14%	86%	0%
Community member	31	26%	74%	0%
Traditional leader	74	11%	86%	3%
District Forestry Officer	50	20%	80%	0%
Private enterprise rep	30	17%	80%	3%

Table 11: Results of the model examining the effect of interviewee role, gender and age on perceived reduction in charcoal production inside the CFA.

Model results for perceived reduction in charcoal production inside the CFA — interviewee role, gender and age

We used a CLMM model random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 248.

Factor	Estimate	SE	*P* value
CFMG executive member	0.829	0.635	0.1919
Traditional leader	0.422	0.636	0.5074
DFO	0.006	0.642	0.9922
Private enterprise rep	0.256	0.712	0.7195
Gender: male	-0.892	0.604	0.1399
Age	-0.001	0.016	0.9361

Charcoal production outside the community forest

To examine for the possibility of leakage, we also asked interviewees whether CFM has led to an increase in charcoal production outside their CFA (Figure 4).

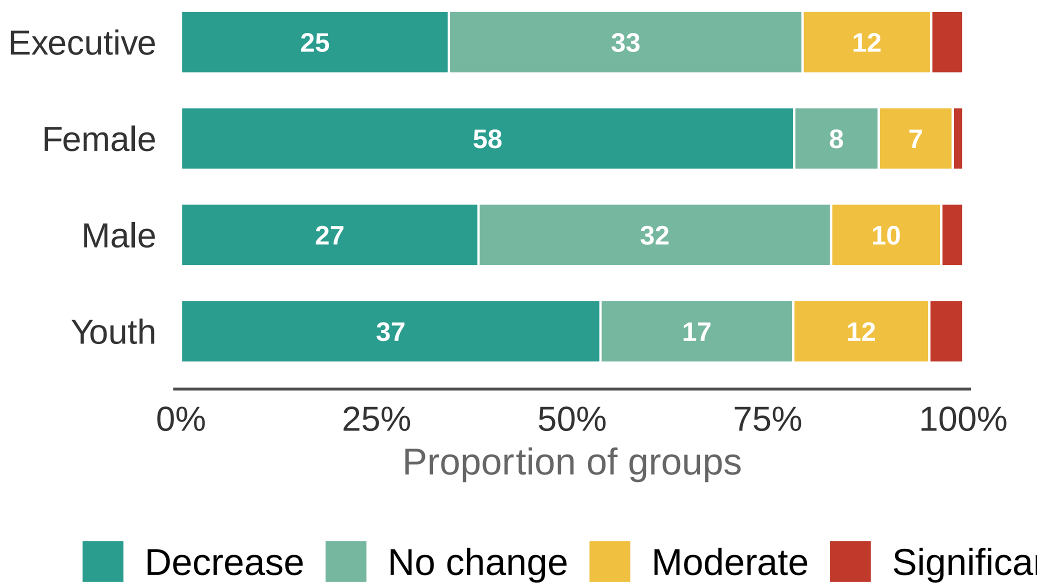

Figure 4: ‘Has CFM increased charcoal production outside the community forest?’ The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen, the vast majority of groups indicated that charcoal making had either decreased outside their CFA or there was no change. However, there were a small number that indicated charcoal making had increased, with some indicating it had increased significantly. Female FGD groups were significantly more likely to indicate that charcoal making had decreased outside the CFA than other groups (Table 12; Table 13). There were no significant differences among these other groups and the effect of the governance index was also not significant.

Among the individual interviewees (Table 14), similar to the results on deforestation, most community interviewees suggested there had been a decrease or no change in charcoal production outside the CFA. However, private enterprise representatives and DFOs were significantly more likely to indicate that there had been an increase in charcoal production outside the CFA (Table 15, Table 16). Interviewee age was also significant — older respondents were more likely to state that there had been an increase in charcoal production outside the CFA.

Table 12: Results of the model examining the effect of FGD type and governance index on perceived increase in charcoal production outside the CFA.

Model results for perceived increase in charcoal production outside the CFA — FGD type and governance index

We used a CLMM with random intercepts for province and CFA within province. Reference = Executive. n = 269.

Factor	Estimate	SE	*P* value
Female	-1.966	0.400	0.0000
Male	-0.055	0.320	0.8627
Youth	-0.566	0.337	0.0931
Governance index	-0.016	0.069	0.8194

Table 13: Pairwise comparisons for perceived increase in charcoal production outside the CFA — FGD type.

Pairwise comparisons for perceived increase in charcoal production outside the CFA — FGD type

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	1.966	0.400	4.919	0.0000
Executive - Male	0.055	0.320	0.173	0.9982
Executive - Youth	0.566	0.337	1.679	0.3346
Female - Male	-1.910	0.409	-4.670	0.0000
Female - Youth	-1.400	0.413	-3.387	0.0039
Male - Youth	0.511	0.346	1.476	0.4521

Table 14: Perceived increase in charcoal production outside the community forest by individual interviewee role.

Perceived increase in charcoal production outside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	No change	Moderate	Significant
CFMG executive member	63	73%	8%	13%	6%
Community member	30	73%	7%	20%	0%
Traditional leader	73	70%	14%	10%	7%
District Forestry Officer	45	40%	20%	27%	13%
Private enterprise rep	30	50%	17%	30%	3%

Table 15: Results of the model examining the effect of interviewee role, gender and age on perceived increase in charcoal production outside the CFA.

Model results for perceived increase in charcoal production outside the CFA — interviewee role, gender and age

We used a CLMM with random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 241.

Factor	Estimate	SE	*P* value
CFMG executive member	0.142	0.598	0.8121
Traditional leader	-0.088	0.573	0.8775
DFO	1.847	0.616	0.0027
Private enterprise rep	1.563	0.651	0.0164
Gender: male	0.093	0.386	0.8101
Age	0.043	0.014	0.0015

Table 16: Pairwise comparisons for perceived increase in charcoal production outside the CFA — interviewee role.

Pairwise comparisons for perceived increase in charcoal production outside the CFA — interviewee role

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Community member - CFMG exec member	-0.142	0.598	-0.238	0.9993
Community member - Trad leader	0.088	0.573	0.154	0.9999
Community member - dfo	-1.847	0.616	-3.001	0.0226
Community member - NGO rep	-1.563	0.651	-2.401	0.1150
CFMG exec member - Trad leader	0.230	0.473	0.487	0.9886
CFMG exec member - dfo	-1.705	0.476	-3.582	0.0031
CFMG exec member - NGO rep	-1.420	0.530	-2.681	0.0567
Trad leader - dfo	-1.936	0.497	-3.895	0.0009
Trad leader - NGO rep	-1.651	0.551	-2.997	0.0228
dfo - NGO rep	0.285	0.477	0.597	0.9756

Fire

We asked interviewees about fire monitoring practices and whether CFM had reduced fire incidence inside their CFA. We did not ask about changes in fire frequency outside the CFA, because clearing fields for agriculture or improving grazing often involves burning.

Fire monitoring

As can be seen in the figure (Figure 5), most CFMGs have instigated a system of fire monitoring. However, ten CFMGs across four provinces from our sample of 75 did not have any system of fire monitoring.

Fire incidence inside the community forest

We asked interviewees whether CFM has reduced fire incidence inside their CFA (Figure 6).

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As can be seen, a majority of groups reported a moderate or significant reduction in fire incidence in the CFA. Female FGDs reported more significant declines in fire frequency compared to the other groups (Table 17). These differences between female FGD groups and the other

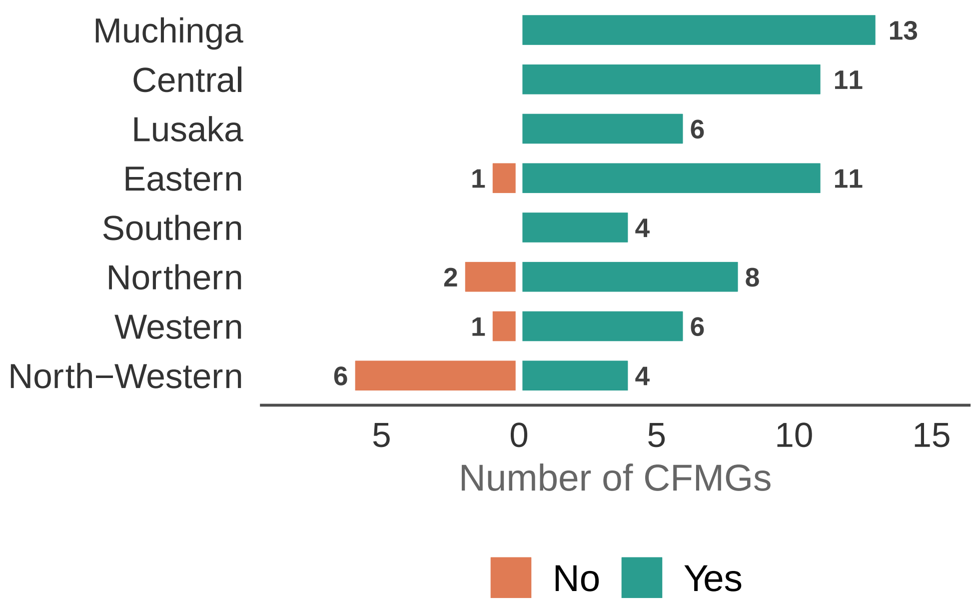

Figure 5: Whether CFMGs have established fire monitoring in the community forest by province. The length of the bars represents the number of CFMGs either monitoring fires (Yes) or not (No). The numbers on the bars represent the actual number of CFMGs in each category.

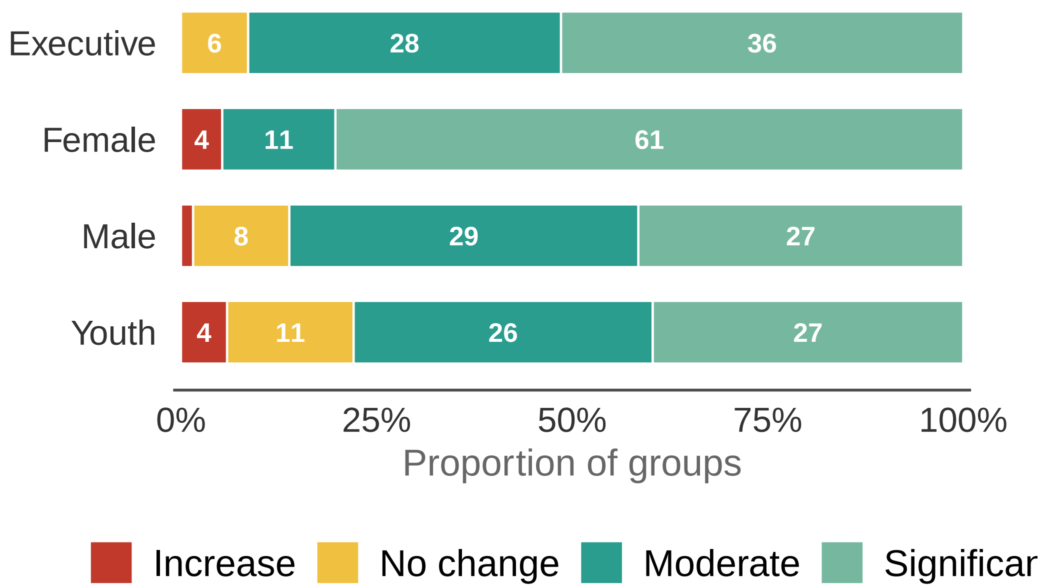

Figure 6: 'Has CFM reduced fire incidence inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 17: Results of the model examining the effect of FGD type and governance index on perceived reduction in fire incidence inside the CFA.

Model results for perceived reduction in fire incidence inside the CFA — FGD type and governance index

We use a CLMM with random intercepts for province and CFA within province. Reference = Executive. n = 261.

Factor	Estimate	SE	*P* value
Female	1.632	0.432	0.0002
Male	-0.460	0.360	0.2010
Youth	-0.717	0.351	0.0410
Governance index	0.167	0.082	0.0422

FGD groups were significant (Table 18). The effect of the governance index was also significant, indicating that CFAs with higher governance scores had greater perceived declines in fire incidence.

Among the individual interviewees, there were also some differences in opinion (Table 19). However, these differences were not significant (Table 20). The effects of interviewee gender and age were also not significant.

Wildlife

We asked interviewees whether CFM had led to an increase in wildlife inside their CFA (Figure 7).

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As can be seen, a majority of groups indicated a significant or moderate increase in wildlife in their CFA. However, there were some that indicated there had been a decrease. The responses among FGD types were not significantly different (Table 21). The effect of governance index was also not significant.

Among the individual interviewees (Table 22), there were some differences in opinion, but again these differences were not significant (Table 23). The effects of interviewee gender and age were also not significant.

Forest resources management

We asked interviewees whether CFM had improved the management of forest resources inside their CFA (Figure 8).

Table 18: Pairwise comparisons for perceived reduction in fire incidence inside the CFA — FGD type.

Pairwise comparisons for perceived reduction in fire incidence inside the CFA — FGD type

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	-1.632	0.432	-3.778	0.0009
Executive - Male	0.460	0.360	1.279	0.5765
Executive - Youth	0.717	0.351	2.044	0.1721
Female - Male	2.092	0.451	4.640	0.0000
Female - Youth	2.349	0.443	5.302	0.0000
Male - Youth	0.257	0.361	0.711	0.8928

Table 19: Perceived reduction in fire incidence inside the community forest by individual interviewee role.

Perceived reduction in fire incidence inside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Significant	Moderate	No change	Increase
CFMG executive member	63	73%	27%	0%	0%
Community member	30	67%	27%	3%	3%
Traditional leader	75	63%	24%	4%	9%
District Forestry Officer	54	44%	39%	17%	0%
Private enterprise rep	31	45%	45%	3%	6%

Table 20: Results of the model examining the effect of interviewee role, gender and age on perceived reduction in fire incidence inside the CFA.

Model results for perceived reduction in fire incidence inside the CFA — interviewee role, gender and age

CLMM; random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 253.

Factor	Estimate	SE	P value
CFMG executive member	0.341	0.514	0.5073
Traditional leader	-0.266	0.514	0.6047
DFO	-0.613	0.520	0.2382
Private enterprise rep	-0.669	0.562	0.2340
Gender: male	-0.609	0.381	0.1104
Age	0.010	0.012	0.4078

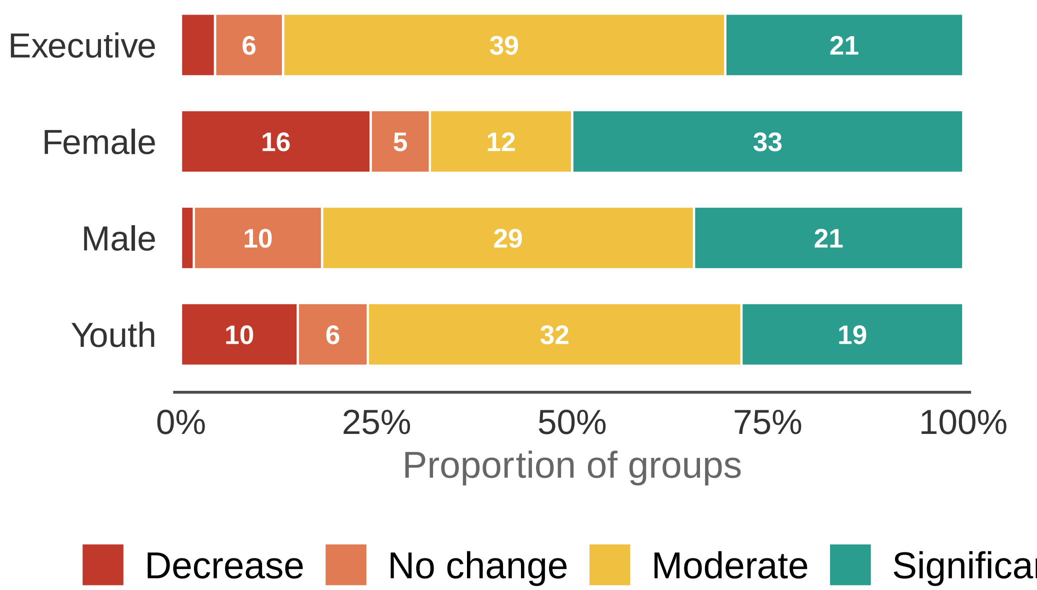

Figure 7: 'Has CFM led to an increase in wildlife in the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 21: Results of the model examining the effect of FGD type and governance index on perceived increase in wildlife inside the CFA.

Model results for perceived increase in wildlife inside the CFA — FGD type and governance index

We used a CLMM with random intercepts for province and CFA within province. Reference = Executive. n = 244.

Factor	Estimate	SE	*P* value
Female	-0.141	0.391	0.7189
Male	-0.174	0.377	0.6442
Youth	-0.382	0.358	0.2861
Governance index	-0.031	0.108	0.7724

Table 22: Perceived increase in wildlife inside the community forest by individual interviewee role.

Perceived increase in wildlife by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	No change	Moderate	Significant
CFMG executive member	61	10%	3%	43%	44%
Community member	27	22%	15%	30%	33%
Traditional leader	73	19%	11%	34%	36%
District Forestry Officer	37	14%	24%	43%	19%
Private enterprise rep	31	3%	10%	48%	39%

Table 23: Results of the model examining the effect of interviewee role, gender and age on perceived increase in wildlife inside the CFA.

Model results for perceived increase in wildlife inside the CFA — interviewee role, gender and age

We used a CLMM with random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 229.

Factor	Estimate	SE	*P* value
CFMG executive member	0.438	0.536	0.4135
Traditional leader	0.105	0.523	0.8408
DFO	-0.499	0.563	0.3753
Private enterprise rep	0.741	0.586	0.2062
Gender: male	-0.663	0.391	0.0899
Age	0.007	0.013	0.6022

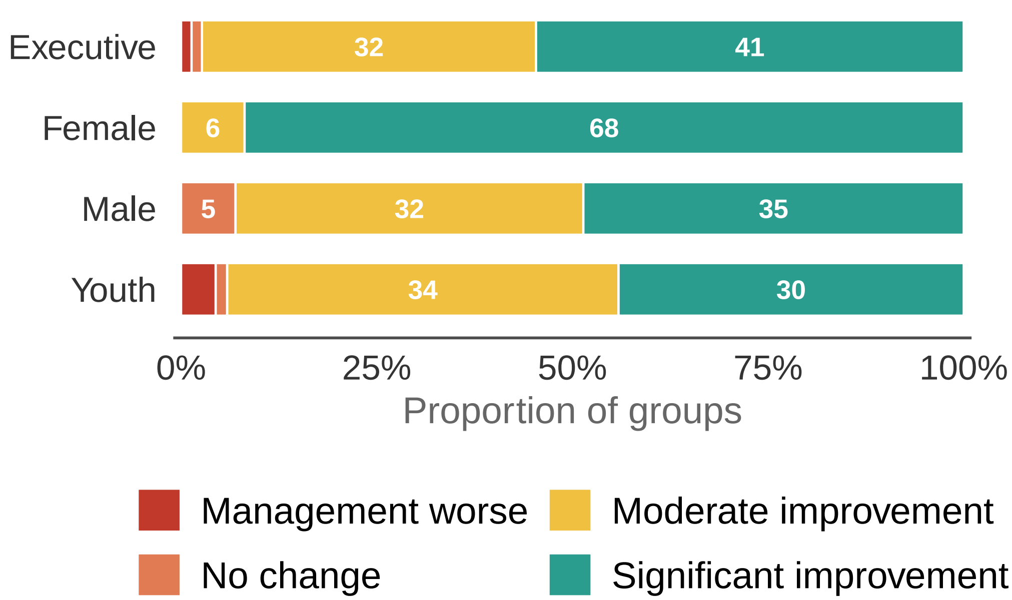

Figure 8: 'Has CFM improved forest resources management inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 24: Results of the model examining the effect of FGD type and governance index on perceived improvement in forest resources management.

Model results for perceived improvement in forest resources management — FGD type and governance index

We used a CLMM with random intercepts for province and CFA within province. Reference = Executive. n = 270.

Factor	Estimate	SE	*P* value
Female	3.379	0.678	0.0000
Male	-0.156	0.389	0.6873
Youth	-0.650	0.387	0.0932
Governance index	0.253	0.110	0.0217

As can be seen, while responses vary among FGD groups, the vast majority of groups agreed that the management of forest resources had improved moderately or significantly under CFM, and only a small number of primarily youth groups considered that it had gotten worse. A very high proportion of community female FGD groups considered that the management of forest resources was significantly better under CFM, a difference that was statistically significant (Table 24, Table 25). The effect of governance index was also significant.

We also asked which specific forest resources had improved under CFM (Figure 9).

Among the individual interviewees (Table 26), the vast majority of interviewees considered that the management of forest resources had improved under CFM. There was a significant difference in responses between community members and DFOs. The DFOs were more likely to suggest there had been no change or only a moderate improvement in forest resource management under CFM, compared to the much higher proportion of community members, CFMG executive members or traditional leaders indicating a substantial improvement in forest resources management (Table 27, Table 28). Interestingly, male interviewees were significantly less likely to perceive improvements in forest resources management (Table 27).

Restoration

We asked CFMG executives whether natural forest restoration is practised inside the community forest (Figure 10). We restricted the responses for this question to CFMG executives, because this is the group that should be best informed about the activities of the CFMG (and the differences in responses among groups to this question are not of interest).

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Table 25: Pairwise comparisons for perceived improvement in forest resources management — FGD type.

Pairwise comparisons for perceived improvement in forest resources management — FGD type

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	-3.379	0.678	-4.987	0.0000
Executive - Male	0.156	0.389	0.403	0.9779
Executive - Youth	0.650	0.387	1.679	0.3348
Female - Male	3.535	0.687	5.144	0.0000
Female - Youth	4.029	0.704	5.722	0.0000
Male - Youth	0.494	0.400	1.233	0.6055

Figure 9: Forest resources that have improved under CFM, as reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting an improvement for each resource.

Table 26: Perceived change in forest resources management inside the community forest by individual interviewee role.

Perceived improvement in forest resources management by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Management worse	No change	Moderate improvement	Significant
CFMG executive member	63	0%	0%	29%	7
Community member	31	3%	0%	32%	6
Traditional leader	75	7%	5%	31%	5
District Forestry Officer	55	4%	11%	55%	3
Private enterprise rep	31	0%	0%	42%	5

Table 27: Results of the model examining the effect of interviewee role, gender and age on perceived improvement in forest resources management.

Model results for perceived improvement in forest resources management — interviewee role, gender and age

We use a CLMM with random intercepts for province and CFA within province. Reference role = Community member; reference gender = female. n = 255.

Factor	Estimate	SE	*P* value
CFMG executive member	0.075	0.595	0.8998
Traditional leader	-0.491	0.561	0.3813
DFO	-1.902	0.593	0.0013
Private enterprise rep	-0.695	0.653	0.2869
Gender: male	-2.333	0.538	0.0000
Age	-0.009	0.014	0.5118

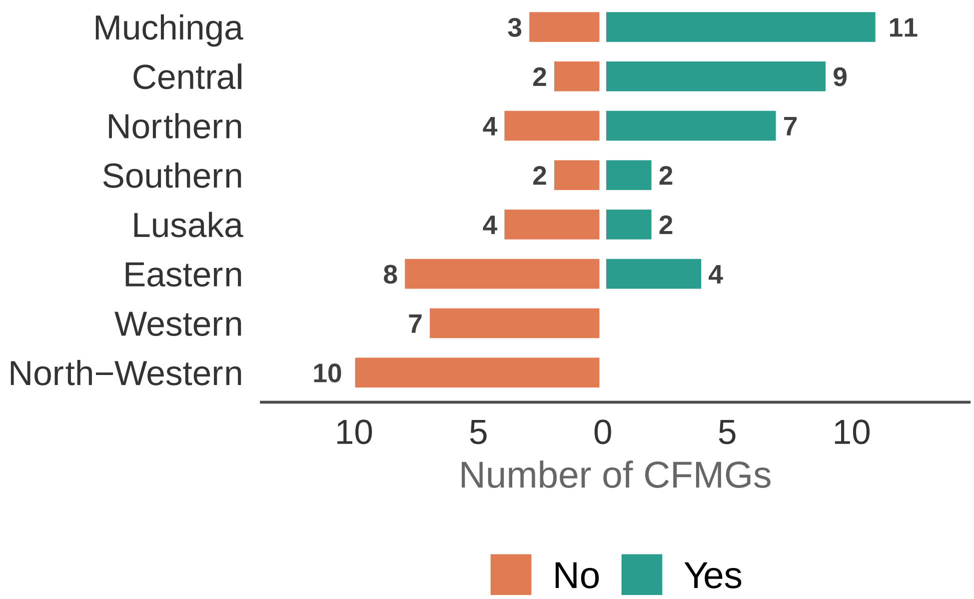

Figure 10: Whether CFMGs undertake natural forest restoration in the community forest by province. The length of the bars represents the number of CFMGs either undertaking restoration (Yes) or not (No). The numbers on the bars represent the actual number of CFMGs in each category.

Table 28: Pairwise comparisons for perceived improvement in forest resources management — interviewee role.

Pairwise comparisons for perceived improvement in forest resources management — interviewee role

Tukey HSD adjustment.

Contrast	Estimate	SE	z ratio	*P* value
Community member - CFMG exec member	-0.075	0.595	-0.126	0.9999
Community member - Trad leader	0.491	0.561	0.875	0.9060
Community member - dfo	1.902	0.593	3.207	0.0117
Community member - NGO rep	0.695	0.653	1.065	0.8246
CFMG exec member - Trad leader	0.566	0.473	1.197	0.7534
CFMG exec member - dfo	1.977	0.484	4.088	0.0004
CFMG exec member - NGO rep	0.770	0.564	1.365	0.6503
Trad leader - dfo	1.411	0.476	2.967	0.0250
Trad leader - NGO rep	0.204	0.563	0.363	0.9963
dfo - NGO rep	-1.207	0.515	-2.344	0.1311

Restoration of natural forests is being practised across approximately half of the CFAs in our sample, mostly in Muchinga, Central and Northern provinces. We also asked CFMG executives about the specific restoration methods applied inside their CFA (Figure 11). The majority were protecting natural regeneration. Only one CFMG was involved in planting native species and removal of invasives was not practised.

Plantations

In addition, we asked CFMG executives whether any plantations had been established inside their CFA (Figure 12). Only seven out the the 75 CFAs were practising any plantation establishment.

We also asked executives to estimate the number of hectares planted in the plantation (Figure 13), but as can bee seen the planted area is very small.

Agroforestry

Last, we asked CFMG executives whether CFM has involved introducing agroforestry practices (Figure 14). In this context, agroforestry is implemented on agricultural land outside the CFA as a means for achieving sustainable intensification and reducing pressure on forests. Around 40% of CFMGs said that agroforestry had been introduced under CFM.

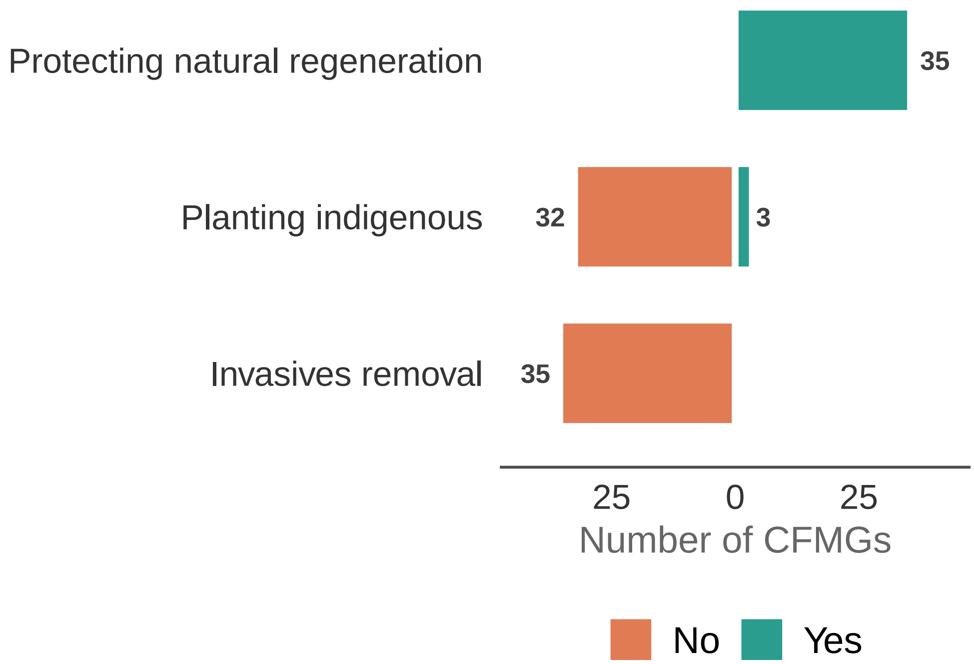

Figure 11: Restoration methods applied inside the community forest, as reported by CFMG executives. The length of the bars represents the number of CFMGs either using (Yes) or not using (No) each method. The numbers on the bars represent the actual number of CFMGs in each category.

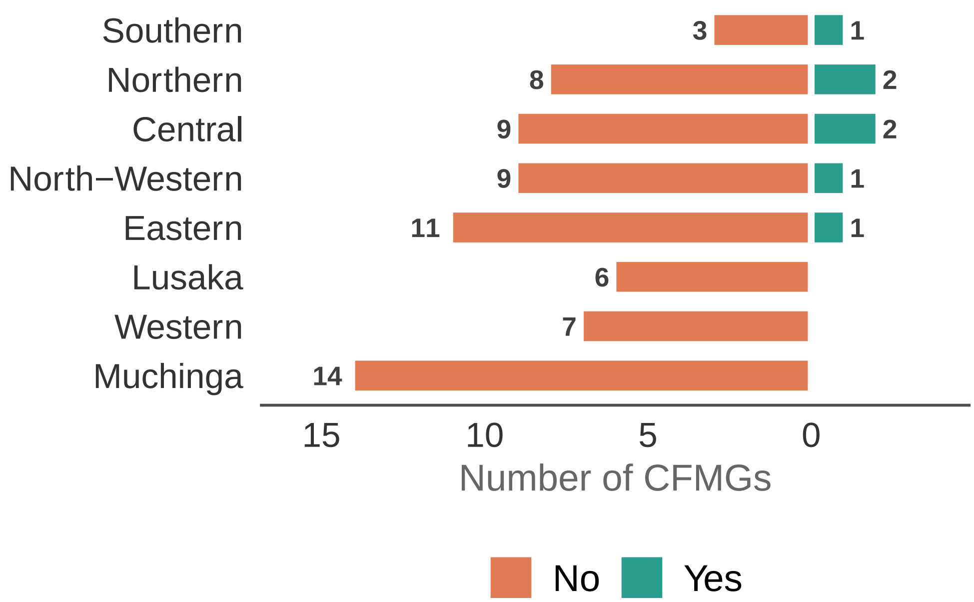

Figure 12: Whether CFMGs have established a plantation inside the community forest, by province. The length of the bars represents the number of CFMGs either with (Yes) or without (No) a plantation. The numbers on the bars represent the actual number of CFMGs in each category.

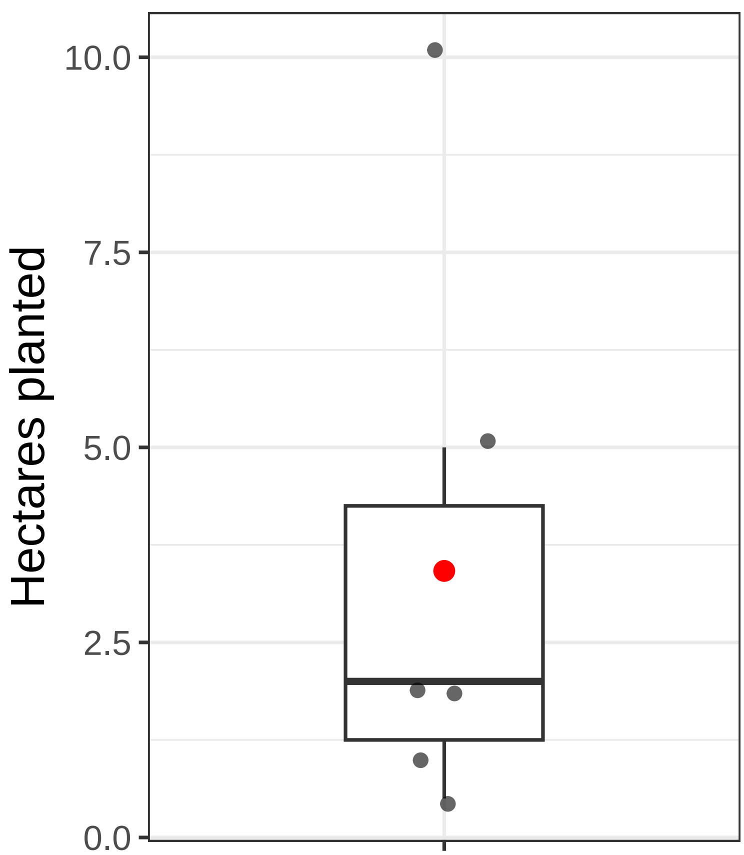

Figure 13: Estimated area of plantation established inside the community forest, as reported by CFMG executives (n = 6 CFMGs; total area = 20.5 ha). The horizontal bar is the median and the red dot is the mean. Species planted include *Gliricidia sepium* (n = 3 CFMGs), *Pinus* spp. (n = 2), *Acacia* spp. (n = 1), *Faidherbia albidia* (n = 1), and miombo species (n = 1).

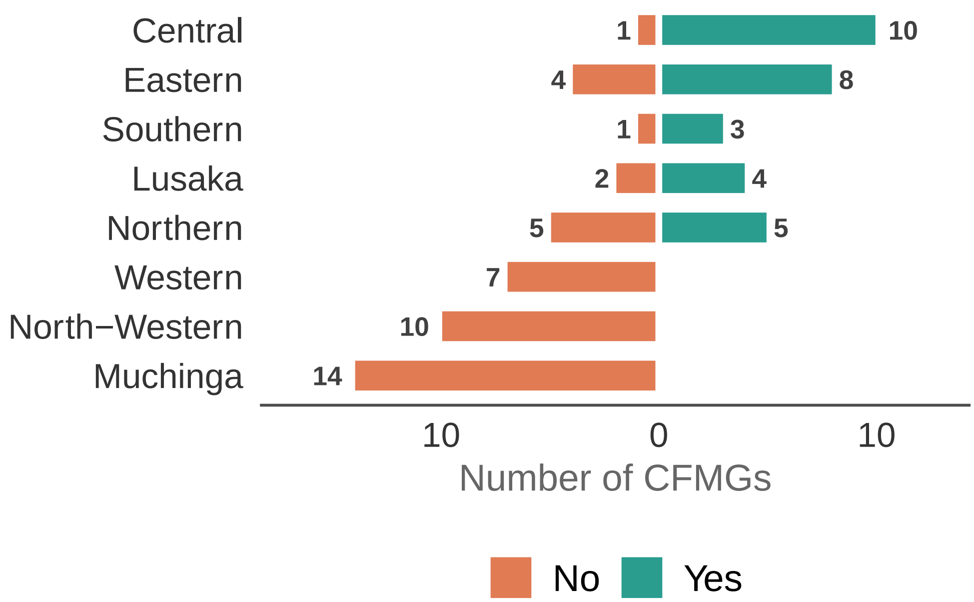

Figure 14: Whether agroforestry is employed to enhance SFM, as reported by CFMG executives. The length of the bars represents the number of CFMGs reporting either a positive impact (Yes) or no impact (No). The numbers on the bars represent the actual number of CFMGs in each category.

We asked CFMG executives about the types of agroforestry practices present in the community (Figure 15).

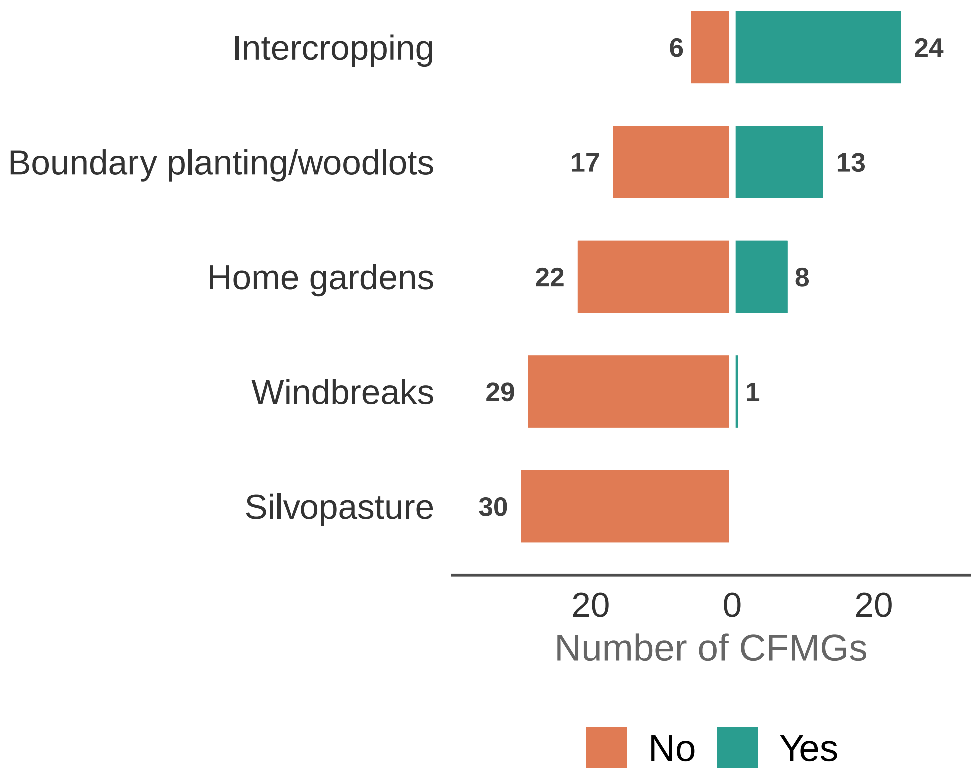

Figure 15: Types of agroforestry practices reported by CFMG executives (n = 30 CFMGs). The length of the bars represents the number of CFMGs either reporting (Yes) or not reporting (No) each practice.

Among the 30 CFMGs practising agroforestry most were employing intercropping with leguminous shrubs for soil health management, but some were planting boundary trees or woodlots and homegardens.

Among the species used (Figure 16), we can see that *Gliricidia* and *Faidherbia*, leguminous shrubs and trees, respectively, that are used for intercropping, were the most popular.

SFM index

To summarise the overall SFM performance of each CFMG, we constructed an SFM score (0–5) by summing five binary indicators of change inside the CFA — decrease in deforestation,

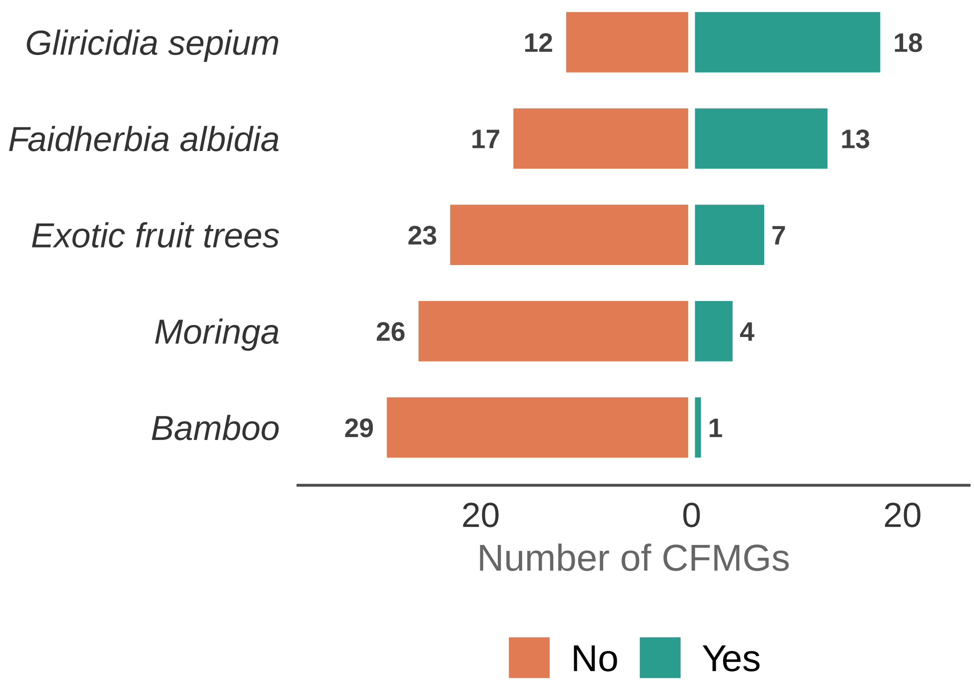

Figure 16: Agroforestry species planted, as reported by CFMG executives (n = 30 CFMGs). The length of the bars represents the number of CFMGs either planting (Yes) or not planting (No) each species.

decrease in charcoal production, decrease in fire incidence, increase in wildlife, and improved natural resources management — scored as 1 (positive change reported) or 0 (no change or worsening). The distribution of SFM scores across CFMGs is shown in Figure 17 (Appendix III). We then modelled this score using a linear mixed model with FGD type and the governance index as predictors, and random intercepts for province and CFA within province, to test whether governance and FGD type were significant predictors of overall SFM performance (Table 29, Figure 18).

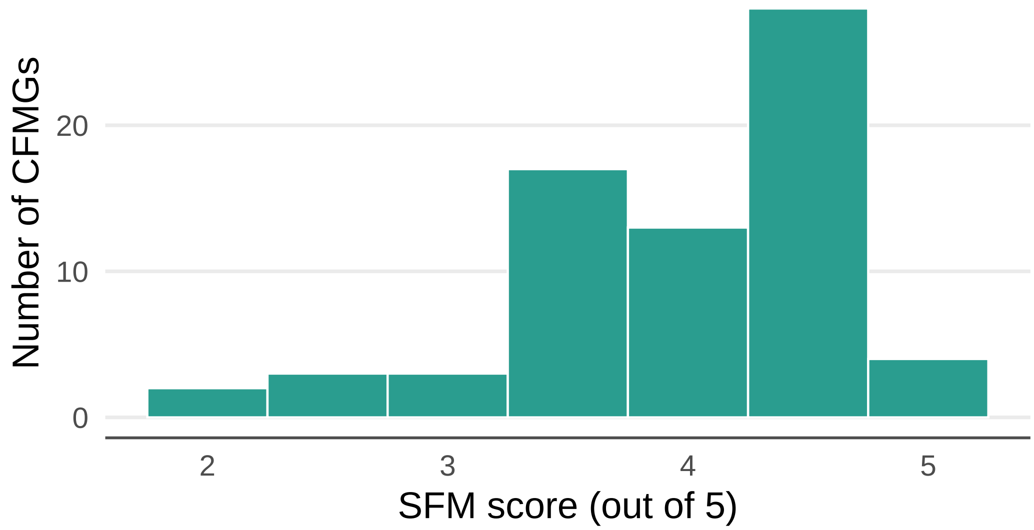

Figure 17: Distribution of mean SFM scores across CFMGs (0–5). Each bar represents the number of CFMGs with that mean score, averaged across all four FGD types. The score counts the number of five SFM indicators showing improvement: reduced deforestation, reduced charcoal production, reduced fire incidence, increased wildlife, and improved natural resource management.

As can be seen from the table of model results, FGD type did not have any significant effect on the SFM index. However, the governance index had a highly significant effect. Thus, CFAs with higher governance index scores (i.e. those that are better governed) had better outcomes for SFM, as reported by the FGDs groups.

Discussion

All the indicators of SFM point to a situation that is improving across that vast majority of CFAs, including deforestation, charcoal production, fire frequency, wildlife abundance and natural resource management. Interestingly, there does not appear to be any indication of leakage for most CFAs, as most FGD groups indicated that both deforestation and charcoal production were also decreasing outside their CFA. In fact, there is some suggestion of positive

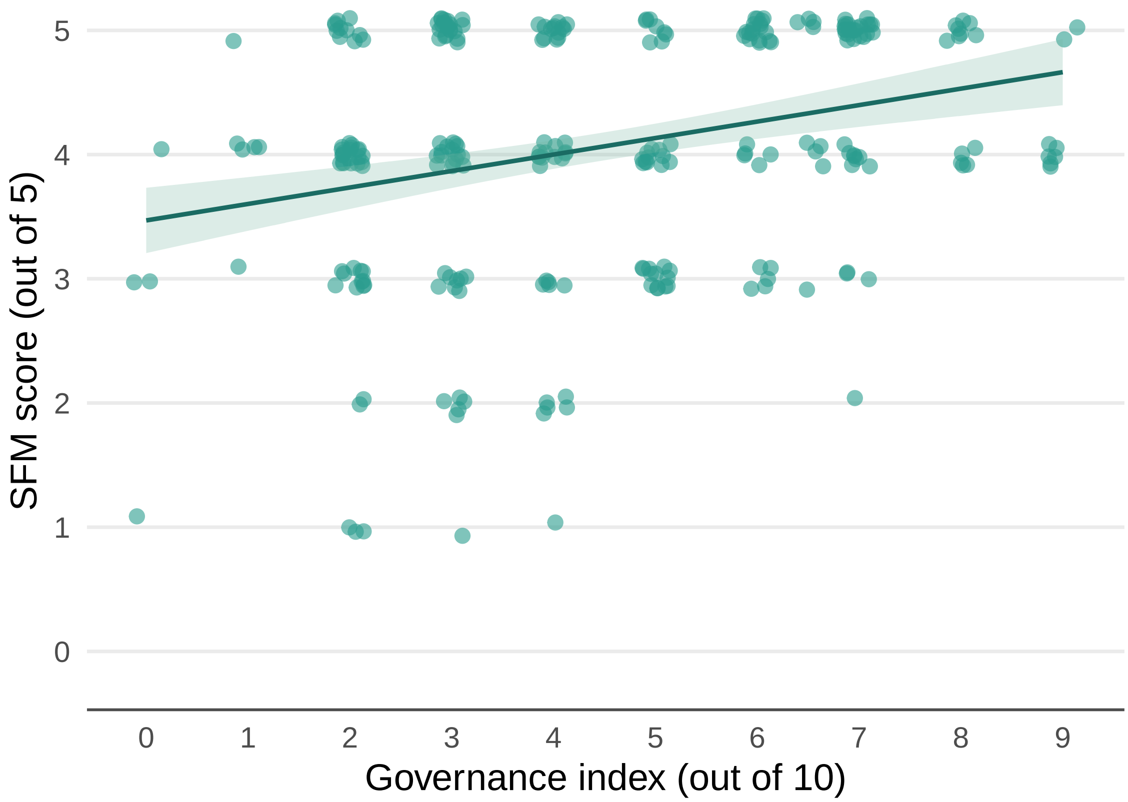

Figure 18: Relationship between the CFMG governance index and SFM score. Each point represents one FGD session. The line shows the fitted relationship from the linear mixed model, with the shaded band representing ± 1 SE.

Table 29: Results of the model examining the effect of FGD type and governance index on SFM performance.

Model results for SFM performance — FGD type and governance index

Linear mixed model with random intercepts for province and CFA within province. Reference = Executive. Province variance = 0.101; CFA-within-province variance = 0.127; residual variance = 0.692. Marginal $R^2 = 0.069$; conditional $R^2 = 0.3$. n = 273.

Factor	Estimate	SE	*P* value
Intercept	3.682	0.217	0.0000
Female	-0.215	0.140	0.1249
Male	-0.113	0.142	0.4303
Youth	-0.201	0.141	0.1550
Governance index	0.115	0.034	0.0011

leakage, whereby the improved management inside CFAs has had a positive effect on their surroundings. There were some differences in responses among the different FGD groups, with community female FGDs tending to have a more positive view of the impact of CFM on SFM compared to the other groups. This is interesting and may indicate that under CFM females have more of a voice in forest and forest resource management. Traditionally, women are highly dependent on forest resources, but usually have little say in forest management decisions. It would be worth following up this study to investigate what is driving this positive take on CFM among community women.

Among the individual interviewees, private enterprise representatives and DFOs tended to have a somewhat more conservative view with respect to positive changes, as compared to individuals from the communities. This may reflect that these professionals have a broader perspective from which to understand changes in forest and forest resource management. However, it could also arise from the fact that they are not as intimately involved in the forest management and hence less aware of the changes. Ultimately, monitoring over a longer period will enable better understanding of how CFM is changing forest management.

Overall, CFM appears to be engendering a change towards improved SFM. However, it is important to note that a small number of FGD groups and individual interviewees suggested the situation was worsening in their CFAs. Moreover, the governance index was a highly significant predictor of the SFM index. For the success of CFM in Zambia, it will be important to identify CFAs that are doing less well, and identify and address the governance issues.

Some CFMGs were practising forest restoration, mostly through protecting natural regeneration, and a few were establishing plantations. However, these activities were not widespread among CFMGs. Restoration - especially through protection of natural regeneration - should integral to CFM and could be implemented across virtually all CFAs in Zambia. As this is an important indicator of SFM and there are government targets to restore 2,000,000 ha by 2030, the FD should integrate restoration targets into forest management plans for CFMGs. A problem is that the boundaries of many existing CFAs were drawn around areas containing relatively good forest (historical deforestation rates are very low in most CFAs — see next chapter on remote sensing). The FD and communities should consider expanding CFA boundaries to incorporate areas for restoration activities and for new CFAs such areas should be incorporated from the outset. Plantations are also an important opportunity for CFM. Plantations can supply local building and fuelwood needs, thereby reducing pressure on natural forests. Furthermore, through sustainable charcoal programmes, they can also supply energy for urban dwellers and again reduce pressure on natural forests. Zambia also has an acute shortage of high quality timber and imports large amounts. A focus on developing plantations through CFM would go some way to meeting these needs, as well as generating a valuable additional source of income for rural communities. As with restoration, the boundaries of CFAs should be planned with plantation activities in mind, which should be focused on heavily degraded area.

In conclusion, our survey data suggest that CFM in Zambia is driving substantial improvements in SFM, reducing deforestation, charcoal production, and fire frequency, increasing wildlife populations, and improving natural resource management. However, it is important to note that these changes were not ubiquitous across all CFAs and CFMGs with lower governance index scores were performing significantly more poorly. While some CFMGs are practising restoration and establishing plantations, there is a substantial opportunity to expand the role of CFM in meeting national restoration and plantation targets.